ANKUSH YADAV

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Education

University of Waterloo — Waterloo, ON, Canada
Masters of Engineering in Electrical and Computer Engineering- Software Specialization — *Sept. 2025 – December 2026*

Rajasthan Techincal University — Rajasthan, India
Bachelor of Technology in Computer Science and Engineering — *May 2019 – June 2023*

Experience

Software Development Engineer Intern — May 2026 – Aug. 2026
Amazon Web Services (AWS) — *Toronto, Canada*

- Built a production security canary that creates fresh database instances every 30 min, executes signed verification scripts on live hosts via a zero-trust transport layer, and auto-raises incidents on failure — cutting detection time for security misconfigurations from days to **under 30 minutes**.
- Engineered a **fault-tolerant dual-path architecture** with **cryptographic multi-party authorization**, **automatic failover**, **retry with backoff**, and session reinit — delivering **100%** verification uptime with **zero operator intervention**.
- Shipped **5 packages**, **2 pipelines**, and full observability(**8 alarms, real-time dashboard, auto-ticketing**) across a multi-account AWS environment using **Java**, **TypeScript**, **CDK**, **CloudWatch**, **Fargate**, and **IAM**.
- Designed for **extensibility at scale**: new database engine teams onboard continuous security monitoring through 2 configuration changes — no new code, infrastructure, or pipelines required.
- Validated security resilience under chaos: **fuzz testing** harness subjects hosts to destructive mutations (**failover**, **reboot**, **network reconfiguration**) and re-verifies all security invariants survive.

Senior Software Developer — Aug. 2023 – April 2026
Alphagrep Securities Pvt Ltd — *Bangalore, India*

- Automated LINUX infrastructure using Python scripting, reducing manual tasks by **24%** and boosting operational efficiency by **16%**.
- Developed a sophisticated application to align database positions with live trading data, reducing expired symbol errors by **29%** and improving data processing speed by **37%**.
- Developed and maintained applications to automate key trading tasks, reducing trader downtime by **11%** and increasing the speed of trade execution by **15%**.
- Designed and implemented MySQL database solutions, improving data storage and retrieval times by **22%**, enabling faster decision-making for traders.
- Built a unified **data ingestion tool** for dumping market data across **multiple exchanges** including NSE, BSE, US, CME, KRX, Binance and China, leveraging the **in-house data team framework**.
- Designed and Developed **C++ binary** for inserting the market data into MySQL database used in **MFT trading infrastructure**.

Web Developer Intern — Jul 2022 – Sep 2022
Vision Freight Solutions India Private Limited — *Jaipur, India*

- Led the development and debugging of assigned projects, delivering high-quality software solutions on time and reducing bug-related issues by **16%**.
- Engineered a dynamic website project, integrating content with a MySQL database, reducing data retrieval times by **18%** and enhancing presentation performance.
- Designed and implemented a secure user login system, increasing system integrity by **24%** and strengthening cybersecurity by ensuring only valid credentials are granted access.

Technical Skills

Languages: Java, Python, C, C++, MySQL, JavaScript, HTML/CSS
Frameworks: React, Node.js, JUnit
Developer Tools: Linux, Jenkins, Git, Docker, Terraform, Ansible, AWS, Google Cloud Platform, VS Code, PyCharm, IntelliJ
Core Competencies: Data Structures and Algorithms , Object Oriented Programming Concepts , Database Management System , Software Development Lifecycle
Libraries: pandas, NumPy, Matplotlib